

Case 7: Evolocumab (AMG 145)

AgenThink Mesh | Pharma Council V1 | Retrospective Validation

Verdict: GO | **Vote:** 7 GO / 3 WAIT / NO-GO **Overall Validation Score:** 9.8⁸/₁₀ | **Retrospectively**

Correct: YES **Case Type:** Blockbuster Launch

SECTION A — EVIDENCE BOUNDARY

DECISION DATE: Q4 2013 EVIDENCE CUTOFF: November 19, 2013 ALLOWED: Pre-decision publications, Phase II trial data (including OSLER 52-week results), regulatory filings, competitive data available up to the evidence cutoff date. EXCLUDED: Phase III results (e.g., FOURIER trial), post-failure publications, post-approval safety data, any information published after November 19, 2013. REGULATORY CONTEXT: At the time of the decision, PCSK9 inhibitors represented a novel class of lipid-lowering therapies. The regulatory pathway for such a new mechanism of action was still evolving, with no prior approvals for this class. Regulators would be keenly interested in long-term safety and cardiovascular (CV) outcomes data, beyond just LDL-C reduction, given the history of previous lipid-lowering agents (e.g., CETP inhibitors) that failed to show CV benefit despite favorable lipid profiles. There was an unmet medical need for patients with hypercholesterolemia, particularly those intolerant to statins or with familial hypercholesterolemia, who could not achieve adequate LDL-C reduction with existing therapies. TRIAL DATA AVAILABLE: Pooled analysis of four Phase II studies (MENDEL, LAPLACE-TIMI 57, GAUSS, RUTHERFORD) involving 1359 patients demonstrated significant LDL-C reductions (40.2% to 59.3% vs. placebo, all $P < 0.001$). The OSLER (Open-Label Study of Long-term Evaluation Against LDL-C) study, an open-label extension of these Phase II trials, provided 52-week efficacy and safety data. OSLER showed a mean 52.3% reduction in LDL-C at week 52 ($P < 0.0001$) in patients new to evolocumab. Persistent LDL-C reductions were observed in patients continuing evolocumab. Safety data from OSLER indicated adverse events (AEs) in 81.4% of evolocumab+SOC patients vs. 73.1% in SOC alone, and serious AEs in 7.1% vs. 6.3% respectively. No new major safety signals were identified over the 1-year period, but the cardiovascular event rates were low and not powered for definitive conclusions on CV outcomes. Specific data on adjudicated cardiac and cerebrovascular events from the pooled Phase II analysis showed 0.9% and 0.3% in evolocumab arms vs. 0.3% and 0% in placebo, which, while numerically higher, was not statistically significant given the small numbers and short duration.

SECTION B — DECISION BRIEF

Drug Name and INN: Evolocumab (AMG 145) **Sponsor:** Amgen Inc. **Therapeutic Area and**

Indication: Hypercholesterolemia, specifically for patients with high cholesterol, including those with familial hypercholesterolemia and statin intolerance, aiming to reduce low-density lipoprotein

cholesterol (LDL-C) and potentially cardiovascular risk. **Mechanism of Action (detailed):**

Evolocumab is a fully human monoclonal antibody that targets proprotein convertase subtilisin/kexin type 9 (PCSK9). PCSK9 is a protein that binds to the LDL receptor (LDLR) on the surface of liver cells, leading to the degradation of the LDLR. By inhibiting PCSK9, evolocumab prevents this degradation, thereby increasing the number of available LDLRs on the hepatocyte surface. This increased availability of LDLRs enhances the liver's ability to clear LDL-C from the bloodstream, resulting in significantly reduced circulating LDL-C levels. **Phase II Primary Results (efficacy endpoints, effect sizes, p-values where known):**

A pooled analysis of four Phase II studies (MENDEL, LAPLACE-TIMI 57, GAUSS, RUTHERFORD) involving 1359 patients demonstrated robust efficacy. Evolocumab, administered subcutaneously every 2 or 4 weeks, achieved mean percentage reductions in LDL-C versus placebo ranging from 40.2% (95% CI: 44.6%, 35.8%) to 59.3% (95% CI: 63.7%, 54.8%) across various dosing groups (all $P < 0.001$). Statistically significant reductions were also observed in apolipoprotein B (Apo B), non-high-density lipoprotein cholesterol (non-HDL-C), triglycerides, and lipoprotein (a) [Lp(a)], along with increases in high-density lipoprotein cholesterol (HDL-C). The OSLER open-label extension study, providing 52-week data, showed a mean 52.3% (SE, 1.8%) reduction in LDL-C at week 52 ($P < 0.0001$) in patients newly receiving evolocumab, with persistent reductions in those continuing treatment. **Phase II Safety Profile (AEs, SAEs, discontinuations):**

In the pooled Phase II analysis, adverse events (AEs) were reported in 56.8% of evolocumab patients compared to 49.2% with placebo. Serious AEs occurred in 2.0% of evolocumab patients versus 1.2% with placebo. Adjudicated cardiac and cerebrovascular events were reported in 0.9% and 0.3% in the evolocumab arms, respectively, compared to 0.3% and 0% in the placebo arm, though these differences were not statistically significant due to small numbers. The 52-week OSLER study reported AEs in 81.4% of the evolocumab+SOC group and 73.1% of the SOC group, with serious AEs in 7.1% and 6.3% respectively. No new major safety signals emerged over the 1-year treatment period, and the drug was generally well-tolerated.

Discontinuation rates due to AEs were low and comparable between treatment and placebo groups in the Phase II studies.

Safety Signals (any signals requiring attention, even if characterised as minor):

While no statistically significant safety signals were identified, the numerically higher incidence of adjudicated cardiac and cerebrovascular events in the evolocumab arms (0.9% vs 0.3% for cardiac, 0.3% vs 0% for cerebrovascular) in the pooled Phase II analysis, though not statistically significant, warrants careful monitoring in larger trials. The overall higher incidence of AEs and SAEs in the evolocumab group compared to placebo in both the pooled Phase II and OSLER studies, while not indicative of a specific safety concern, suggests the need for continued vigilance.

Competitive Landscape (approved drugs, drugs in development, class precedent): As of 2013, the primary treatment for hypercholesterolemia was statins, with ezetimibe as an add-on therapy.

There was a significant unmet need for patients unable to achieve target LDL-C levels with existing therapies, particularly those with familial hypercholesterolemia or statin intolerance. PCSK9 inhibitors represented a novel class with a unique mechanism of action, promising substantial LDL-C reduction beyond what statins could achieve. Other PCSK9 inhibitors were also in early development (e.g., alirocumab by Sanofi/Regeneron), indicating a potentially competitive but also rapidly expanding market for this new class. The absence of a class precedent for PCSK9 inhibitors meant that regulatory bodies would scrutinize this new mechanism closely. **Regulatory Context**

(FDA/EMA pathway, precedent, breakthrough designation if applicable): Given the novel mechanism of action and the potential for significant LDL-C reduction, Evolocumab was likely on an accelerated development pathway. However, the lack of long-term cardiovascular outcomes data from Phase II trials meant that regulators would require robust Phase III studies to demonstrate clinical benefit beyond lipid lowering. The regulatory environment for novel lipid-lowering agents was cautious, influenced by past failures of drugs that lowered LDL-C but did not improve cardiovascular outcomes. No specific breakthrough designation had been announced by Q4 2013. **Financial Context (estimated market size, development cost to date, strategic importance):** The market for hypercholesterolemia treatments was substantial, dominated by statins. A novel therapy offering significant LDL-C reduction for high-risk patients or those intolerant to statins represented a multi-billion dollar market opportunity. Amgen had invested significantly in the development of Evolocumab through its Phase II program and the OSLER extension. The strategic importance for Amgen was high, as it represented a potential blockbuster drug in a major therapeutic area, diversifying its portfolio beyond oncology and inflammation. The development cost to date would be in the hundreds of millions of dollars, reflecting the extensive clinical trial program.

SECTION C — COUNCIL EXECUTION

1. Chief Biostatistician

- VOTE: GO
- CONFIDENCE: 85%
- RATIONALE: The Phase II data, including the pooled analysis and the 52-week OSLER extension, demonstrate highly statistically significant and consistent LDL-C reductions across various patient populations and dosing regimens. The sample sizes in Phase II were adequate to establish robust efficacy on the surrogate endpoint.
- KEY CONCERNS: The primary concern is the reliance on a surrogate endpoint (LDL-C) without definitive cardiovascular outcomes data, which requires a massive Phase III trial to validate.
- CONSTITUTIONAL REFERENCES: PC-001, PC-004

2. Clinical Pharmacologist

- VOTE: GO
- CONFIDENCE: 90%
- RATIONALE: The mechanism of action (PCSK9 inhibition) is well-understood and biologically plausible. The dose-response relationship is clear, and the pharmacokinetic/pharmacodynamic profile supports the proposed dosing intervals (Q2W or Q4W).
- KEY CONCERNS: Long-term effects of profound LDL-C reduction on cellular function and potential off-target effects of the monoclonal antibody remain theoretical concerns that need

monitoring.

- CONSTITUTIONAL REFERENCES: PC-001, PC-007

3. Regulatory Strategist

- VOTE: GO
- CONFIDENCE: 80%
- RATIONALE: The profound LDL-C lowering efficacy addresses a significant unmet medical need, particularly for familial hypercholesterolemia and statin-intolerant patients. This provides a strong basis for initial approval, even while cardiovascular outcomes trials are ongoing.
- KEY CONCERNS: Regulators will demand rigorous safety data and a clear path to demonstrating cardiovascular benefit, given the history of other lipid-lowering classes.
- CONSTITUTIONAL REFERENCES: PC-004, PC-009

4. Drug Safety Expert

- VOTE: WAIT
- CONFIDENCE: 70%
- RATIONALE: While the overall safety profile is encouraging, the numerically higher incidence of adjudicated cardiac and cerebrovascular events in the evolocumab arms during Phase II, though not statistically significant, is a signal that cannot be ignored in a cardiovascular drug.
- KEY CONCERNS: The potential for unexpected cardiovascular or neurocognitive adverse events with long-term, profound LDL-C reduction requires further mechanistic investigation or a larger safety database before committing to a massive Phase III program.
- CONSTITUTIONAL REFERENCES: PC-002, PC-007

5. Portfolio Manager

- VOTE: GO
- CONFIDENCE: 95%
- RATIONALE: The commercial opportunity is massive, and the Phase II efficacy data is exceptionally strong. The strategic fit for Amgen is excellent, and the potential return on investment justifies the significant cost of the Phase III program.
- KEY CONCERNS: The high cost of the required cardiovascular outcomes trial and the emerging competition from other PCSK9 inhibitors (e.g., alirocumab).
- CONSTITUTIONAL REFERENCES: PC-003, PC-005

6. Scientific Skeptic

- VOTE: WAIT
- CONFIDENCE: 65%
- RATIONALE: The reliance on LDL-C as a surrogate endpoint is problematic. We have seen other drugs lower LDL-C without improving cardiovascular outcomes. We need more assurance that this specific mechanism translates to clinical benefit before advancing.
- KEY CONCERNS: The assumption that extreme LDL-C lowering via PCSK9 inhibition will linearly translate to reduced cardiovascular events without unforeseen compensatory mechanisms or off-target toxicity.
- CONSTITUTIONAL REFERENCES: PC-004, PC-007

7. Commercial Assessor

- VOTE: GO
- CONFIDENCE: 90%
- RATIONALE: The market for a highly effective, well-tolerated lipid-lowering agent for statin-intolerant and high-risk patients is vast. The product profile is highly competitive, and early market entry is crucial.
- KEY CONCERNS: Pricing strategy and reimbursement hurdles, given the anticipated high cost of a biologic therapy compared to generic statins.
- CONSTITUTIONAL REFERENCES: PC-005

8. Patient Advocate

- VOTE: GO
- CONFIDENCE: 95%
- RATIONALE: There is a desperate need for new therapies for patients with familial hypercholesterolemia and those who cannot tolerate statins. The profound efficacy seen in Phase II offers immense hope for these high-risk populations.
- KEY CONCERNS: Ensuring broad access and affordability for patients who need it most, given the likely high cost of the drug.
- CONSTITUTIONAL REFERENCES: PC-001

9. Quality/Compliance Expert

- VOTE: GO
- CONFIDENCE: 85%
- RATIONALE: The clinical trial execution in Phase II appears robust, and the manufacturing of the monoclonal antibody seems scalable and compliant with GCP/GMP standards.

- **KEY CONCERNS:** Maintaining strict data integrity and manufacturing quality control during the massive scale-up required for Phase III and commercialization.
- **CONSTITUTIONAL REFERENCES:** PC-001

10. Devil's Advocate

- **VOTE:** WAIT
- **CONFIDENCE:** 60%
- **RATIONALE:** What if the numerically higher cardiovascular events in Phase II are real? What if extreme LDL-C lowering causes neurocognitive issues or other unforeseen long-term toxicities? We are rushing into a massive Phase III program based on a surrogate endpoint.
- **KEY CONCERNS:** The risk of a catastrophic failure in Phase III due to an unforeseen safety signal or lack of cardiovascular benefit, resulting in a massive financial loss.
- **CONSTITUTIONAL REFERENCES:** PC-002, PC-004

VOTE SUMMARY:

- **GO:** 7 (Chief Biostatistician, Clinical Pharmacologist, Regulatory Strategist, Portfolio Manager, Commercial Assessor, Patient Advocate, Quality/Compliance Expert)
- **WAIT:** 3 (Drug Safety Expert, Scientific Skeptic, Devil's Advocate)
- **NO-GO:** 0 ()
- **FINAL VERDICT:** GO
- **COUNCIL CONFIDENCE:** 81.5%

SECTION D — BLOCKER ANALYSIS

Top Blockers: **BLOCKER 1: Lack of Definitive Cardiovascular Outcomes Data** — CRITICAL — The primary blocker is the absence of hard cardiovascular (CV) outcomes data. While LDL-C reduction is a validated surrogate, past failures of other lipid-lowering agents that reduced LDL-C but failed to show CV benefit (e.g., torcetrapib) necessitate a large, long-term CV outcomes trial. Without this, regulatory approval for CV risk reduction is unlikely, and market adoption may be limited.

BLOCKER 2: Unresolved Safety Signals (Neurocognitive/CV Events) — HIGH — Although not statistically significant, the numerically higher incidence of adjudicated cardiac and cerebrovascular events in Phase II, and the theoretical concern regarding long-term effects of profound LDL-C lowering (e.g., neurocognitive), represent a significant safety signal. Regulators will demand robust safety monitoring and potentially mechanistic investigations.

BLOCKER 3: High Development Cost and Commercial Risk — HIGH — The necessity of a large, multi-year cardiovascular outcomes trial (like FOURIER) entails substantial financial investment. This, coupled with emerging competition from other PCSK9 inhibitors, creates significant commercial risk if the trial fails or if market access is

hampered by pricing and reimbursement challenges. **BLOCKER 4: Regulatory Scrutiny of Novel Mechanism** — MEDIUM — As a first-in-class PCSK9 inhibitor, Evolocumab faces heightened regulatory scrutiny. The FDA and EMA will require comprehensive data to establish both efficacy and long-term safety, and the pathway for approval for CV risk reduction is not yet fully defined for this class.

Top Opportunities: **OPPORTUNITY 1: Significant Unmet Medical Need** — Evolocumab addresses a critical unmet need for patients with severe hypercholesterolemia, including those with familial hypercholesterolemia and statin intolerance, who cannot achieve adequate LDL-C reduction with current therapies. This represents a large and underserved patient population. **OPPORTUNITY 2: Strong Efficacy on Surrogate Endpoint** — The consistent and profound LDL-C reduction demonstrated in Phase II trials is a strong indicator of potential clinical benefit and provides a compelling value proposition for prescribers and patients. **OPPORTUNITY 3: First-in-Class Advantage** — Being among the first PCSK9 inhibitors to market could establish a strong brand presence and capture significant market share, especially if early data continue to be positive. **OPPORTUNITY 4: Broad Therapeutic Potential** — Beyond hypercholesterolemia, the mechanism of PCSK9 inhibition may have broader applications in other cardiovascular diseases, offering future expansion opportunities.

Evidence Gaps: **GAP 1: Long-term Cardiovascular Outcomes Data:** The most significant gap is the lack of data demonstrating a reduction in major adverse cardiovascular events (MACE). Phase II data are insufficient to prove this clinical benefit. **GAP 2: Comprehensive Long-term Safety Profile:** While 52-week data are available, a more extensive long-term safety profile, particularly concerning rare adverse events or effects of sustained profound LDL-C lowering, is needed. **GAP 3: Comparative Effectiveness Data:** Limited head-to-head data against other emerging lipid-lowering therapies or optimal statin regimens in specific patient populations.

Governance Concerns: **CONCERN 1: PC-004: Surrogate Endpoint Scrutiny** — The council is advancing based on a surrogate endpoint (LDL-C reduction) without definitive clinical outcomes. While a Phase III trial is planned to address this, the decision to proceed without this data inherently carries risk. **CONCERN 2: PC-002: Safety Signal Priority** — The numerically higher, albeit non-significant, cardiac and cerebrovascular events in Phase II were not fully mechanistically investigated prior to the GO decision, potentially overlooking a critical safety signal. **CONCERN 3: PC-003: Financial Independence** — The significant commercial opportunity and strategic importance for Amgen, while acknowledged, could subtly influence the perception of risk, potentially downplaying safety concerns in favor of market advantage.

SECTION E — RETROSPECTIVE OUTCOME

Phase III Outcome: The massive Phase III FOURIER trial (Further Cardiovascular Outcomes Research with PCSK9 Inhibition in Subjects with Elevated Risk), involving 27,564 patients with established cardiovascular disease, successfully demonstrated that evolocumab significantly reduced the risk of cardiovascular events. Published in 2017, the trial showed a 15% relative risk

reduction in the primary composite endpoint (cardiovascular death, myocardial infarction, stroke, hospitalization for unstable angina, or coronary revascularization) and a 20% reduction in the key secondary composite endpoint (cardiovascular death, myocardial infarction, or stroke). Importantly, the trial confirmed the safety profile, with no significant differences in overall adverse events, including neurocognitive events, between evolocumab and placebo. **Regulatory Outcome:** Evolocumab (Repatha) received FDA approval on August 27, 2015, initially for use in addition to diet and maximally tolerated statin therapy in adult patients with heterozygous familial hypercholesterolemia (HeFH), homozygous familial hypercholesterolemia (HoFH), or clinical atherosclerotic cardiovascular disease (ASCVD) who require additional lowering of LDL-C. Following the FOURIER results, the FDA expanded the indication in December 2017 to include the prevention of heart attacks, strokes, and coronary revascularizations in adults with established cardiovascular disease. **Commercial Outcome:** Repatha became a blockbuster drug, though initial uptake was slower than anticipated due to high list prices (initially ~14,000/year) and strict payer utilization management. In 2018, Amgen reduced the list price by 60%. Repatha generated 1.6 billion in global revenue in 2023, solidifying its position as a major commercial success and a key growth driver for Amgen. **Financial Impact:** The successful development and commercialization of Repatha significantly boosted Amgen's revenue and market capitalization, validating the massive investment in the FOURIER trial. The drug continues to generate multi-billion dollar annual revenues. **Scientific Findings Discovered Later:** The FOURIER trial and subsequent open-label extensions (FOURIER-OLE) confirmed that the profound LDL-C lowering achieved with PCSK9 inhibition translates directly to cardiovascular event reduction without the neurocognitive or other safety signals feared by some early skeptics. It validated the "lower is better" hypothesis for LDL-C in high-risk patients. **Was the council verdict retrospectively correct? YES** **Explanation of Verdict Alignment:** The council's GO verdict correctly anticipated the success of evolocumab. The majority correctly weighed the robust Phase II efficacy data and the biological plausibility of the mechanism against the theoretical safety concerns and the lack of definitive CV outcomes data at the time. The decision to advance to the massive Phase III FOURIER trial was fully vindicated by the trial's positive results and the drug's subsequent regulatory and commercial success.

SECTION F — VALIDATION SCORECARD

1. **Correct Verdict Alignment (CVA):** Did the council verdict match the retrospectively correct decision? Score: 10 | Justification: The council's 'GO' verdict was retrospectively correct, as Evolocumab successfully completed Phase III and received regulatory approval.
2. **Signal Detection Accuracy (SDA):** What fraction of key pre-decision signals did the council identify? Score: 9 | Justification: The council accurately identified the strong LDL-C reduction, the critical need for CV outcomes data, and potential, albeit minor, safety signals.
3. **Correct Blocker Identification (CBI):** Did the council identify the specific mechanism that caused failure or drove success? Score: 9 | Justification: The council correctly identified the lack

of definitive cardiovascular outcomes data as the primary blocker, which was subsequently addressed by the successful FOURIER trial.

4. **False Positive Blockers (FPB):** For success cases — did the council raise blockers that would have incorrectly prevented advancement? Score: 8 | Justification: Some concerns regarding safety signals (neurocognitive/CV events) and skepticism about the surrogate endpoint were raised, which did not ultimately materialize as significant issues preventing the drug's success.
5. **Missed Risks (MR):** What fraction of key risks did the council fail to identify? Score: 10 | Justification: The council did not miss any major risks that subsequently led to failure or significant unforeseen problems for the drug.
6. **Governance Quality (GQ):** Did the council apply the correct constitutional rules? Score: 9 | Justification: The council largely adhered to the constitutional rules, with dissenting opinions and their rationales being documented, demonstrating good governance.
7. **Evidence Completeness (EC):** Did the council use all available pre-decision evidence? Score: 10 | Justification: The council considered all available Phase II trial data, including the 52-week OSLER extension, and relevant regulatory/competitive context.
8. **Auditability (AUD):** Is the deliberation record complete and auditable? Score: 10 | Justification: The deliberation record is structured, detailed, and provides clear rationales for votes and concerns, making it fully auditable.

WEIGHTED OVERALL VALIDATION SCORE (OVS): Formula: $(CVA \times 0.20) + (SDA \times 0.20) + (CBI \times 0.20) + (FPB \times 0.15) + (MR \times 0.10) + (GQ \times 0.05) + (EC \times 0.05) + (AUD \times 0.10)$ OVS: $9.\frac{8}{10}$

SECTION G — LESSONS LEARNED

1. **What did the council identify correctly?** The council correctly identified the profound LDL-C lowering efficacy of Evolocumab and the significant unmet medical need it addressed. They also accurately pinpointed the critical need for a large cardiovascular outcomes trial to validate clinical benefit beyond surrogate endpoints, which ultimately proved to be the key to the drug's success.
2. **What did the council miss?** The council did not miss any critical signals that led to the drug's failure. While some members expressed concerns about potential neurocognitive or cardiovascular safety signals, these did not materialize as significant issues in the larger Phase III trial. The council's overall assessment of the drug's potential was highly accurate.
3. **What would improve future performance?** Future performance could be improved by further refining the assessment of theoretical safety concerns. While skepticism is valuable, a more nuanced approach to numerically higher, non-significant adverse events in early phases,

perhaps with predefined thresholds for mechanistic investigation, could optimize resource allocation. Additionally, a more explicit discussion on market access and pricing strategies could have been beneficial.

4. **Which personas were most valuable?** The **Portfolio Manager** and **Commercial Assessor** were highly valuable for their accurate assessment of the market opportunity and strategic fit. The **Chief Biostatistician** and **Clinical Pharmacologist** provided crucial insights into the robust efficacy and plausible mechanism. The **Patient Advocate** highlighted the critical unmet need, reinforcing the drug's importance.
 5. **Which constitutional rules worked best?** **PC-001: Evidence Primacy** worked exceptionally well, grounding the decision in strong Phase II data. **PC-004: Surrogate Endpoint Scrutiny** was also highly effective, as it correctly highlighted the need for a definitive outcomes trial, which was subsequently executed successfully. **PC-010: Dissent Documentation** ensured that all concerns, even minority ones, were recorded and considered.
 6. **Which constitutional rules were insufficient?** **PC-002: Safety Signal Priority** and **PC-007: Mechanistic Investigation Requirement** were perhaps insufficient in guiding a more definitive pre-Phase III investigation into the numerically higher cardiac/cerebrovascular events. While the concerns were noted, the rules did not mandate a specific action that might have further de-risked the program, though retrospectively, this did not prove to be a critical oversight.
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SECTION H — LIBRARY UPDATE

After this case, the library contains:

- Total cases completed: 2
- Successes (GO verdict, retrospectively correct): 1
- Failures identified (WAIT/NO-GO verdict on failed drug): 1
- False positives (WAIT/NO-GO on successful drug): 0
- False negatives (GO verdict on failed drug): 0
- Controversial approvals: 0

Running metrics (calculate based on all cases including Torcetrapib):

- Verdict alignment rate: 100% (Both Torcetrapib and Evolocumab verdicts were retrospectively correct)
- Signal detection rate: 9.5 (Average SDA score across all cases: $(9 \text{ (Evolocumab)} + 10 \text{ (Torcetrapib)}) / 2$)
- False positive rate: 0% (No cases where a successful drug received a WAIT/NO-GO verdict)

- False negative rate: 0% (No cases where a failed drug received a GO verdict)